COASTAL LANDFORMS / INDIA COAST AND CRZ

COASTAL-SYSTEM BALANCE → REFRACTION MAP → EROSIONAL SEQUENCE → CLIFF RETREAT SECTION → DEPOSITIONAL DISCRIMINATOR → RELATIVE SEA-LEVEL FORK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW · ● SUBORDINATE EXTRA · ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 COASTAL-SYSTEM BALANCE

COASTAL-SYSTEM BALANCE · WAVES + TIDES + CURRENTS · MARINE ENERGY · RIVERS + CLIFFS + OFFSHORE · SEDIMENT SUPPLY · SEA LEVEL + LAND MOTION · ACCOMMODATION SPACE · HUMAN WORKS

WAVES + TIDES + CURRENTS → marine energy → RIVERS + CLIFFS + OFFSHORE → sediment supply → SEA LEVEL + LAND MOTION → accommodation space → HUMAN WORKS → redistribute erosion and accretion → MUST REMEMBER: Coastal morphology connects wave generation, refraction, erosion, longshore drift, sediment cells, deposition and

CORE CLAIM

- WAVES + TIDES + CURRENTS → marine energy
- RIVERS + CLIFFS + OFFSHORE → sediment supply

MECHANISM

- SEA LEVEL + LAND MOTION → accommodation space
- HUMAN WORKS → redistribute erosion and accretion

CONSEQUENCE / LIMIT

- MUST REMEMBER: Coastal morphology connects wave generation, refraction, erosion, longshore drift, sediment cells, deposition and
- sea-level/tectonic change to cliffs, caves-arches-stacks, beaches, spits, bars, lagoons, deltas and India's east-west coast contrast.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Coastal morphology connects wave generation, refraction, erosion, longshore drift, sediment cells, deposition and sea-level/tectonic change to cliffs, caves-arches-stacks, beaches, spits, bars, lagoons, deltas and India's east-west coast contrast.

01

STAGE 01 REFRACTION MAP

REFRACTION MAP · DEEP WATER · WAVE FRONTS TRAVEL FASTER · SHALLOW HEADLAND · FRONTS SLOW AND CONVERGE · FRONTS DIVERGE AND ENERGY DISPERSES · EROSION FOCUS · DEPOSITIONAL SHELTER

DEEP WATER → wave fronts travel faster

SHALLOW HEADLAND → fronts slow and converge → BAY → fronts diverge and energy disperses → RESULT → erosion focus / depositional shelter

CORE CLAIM

- DEEP WATER → wave fronts travel faster
- SHALLOW HEADLAND → fronts slow and converge

MECHANISM

- BAY → fronts diverge and energy disperses
- RESULT → erosion focus / depositional shelter

CONSEQUENCE / LIMIT

- DEEP WATER → wave fronts travel faster
- SHALLOW HEADLAND → fronts slow and converge

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BAY → fronts diverge and energy disperses; RESULT → erosion focus / depositional shelter.

02

STAGE 02 EROSIONAL SEQUENCE

EROSIONAL SEQUENCE · HYDRAULIC ACTION + ABRASION · WIDENING AND ROOF ATTACK · ROOF COLLAPSE

JOINT / FAULT → hydraulic action + abrasion → CAVE → widening and roof attack → ARCH → roof collapse → STACK → STACK → undercutting → STUMP

CORE CLAIM

- JOINT / FAULT → hydraulic action + abrasion
- CAVE → widening and roof attack

MECHANISM

- ARCH → roof collapse → STACK
- STACK → undercutting → STUMP

CONSEQUENCE / LIMIT

- JOINT / FAULT → hydraulic action + abrasion
- CAVE → widening and roof attack

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: JOINT / FAULT → hydraulic action + abrasion.

03

STAGE 03 CLIFF RETREAT SECTION

CLIFF RETREAT SECTION · WAVE NOTCH · BASAL UNDERCUTTING · INSTABILITY AND COLLAPSE · DEBRIS REMOVAL · RENEWED ATTACK · WIDENING WAVE-CUT PLATFORM

CORE CLAIM

- WAVE NOTCH → basal undercutting
- OVERHANG → instability and collapse

MECHANISM

- DEBRIS REMOVAL → renewed attack
- RETREAT → widening wave-cut platform

CONSEQUENCE / LIMIT

- WAVE NOTCH → basal undercutting
- OVERHANG → instability and collapse

• JOINT / FAULT → hydraulic action + abrasion
• CAVE → widening and roof attack

• ARCH → roof collapse → STACK
• STACK → undercutting → STUMP

• JOINT / FAULT → hydraulic action + abrasion
• CAVE → widening and roof attack

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: JOINT / FAULT → hydraulic action + abrasion.

03

STAGE 03

CLIFF RETREAT SECTION

CLIFF RETREAT SECTION · WAVE NOTCH · BASAL UNDERCUTTING · INSTABILITY AND COLLAPSE · DEBRIS REMOVAL · RENEWED ATTACK · WIDENING WAVE-CUT PLATFORM

CORE CLAIM
• WAVE NOTCH → basal undercutting
• OVERHANG → instability and collapse

MECHANISM
• DEBRIS REMOVAL → renewed attack
• RETREAT → widening wave-cut platform

CONSEQUENCE / LIMIT
• WAVE NOTCH → basal undercutting
• OVERHANG → instability and collapse

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DEBRIS REMOVAL → renewed attack; RETREAT → widening wave-cut platform.

04

STAGE 04

DEPOSITIONAL DISCRIMINATOR

DEPOSITIONAL DISCRIMINATOR · ONE END ATTACHED · CROSSES BAY OR MOUTH · WATER ENCLOSED BEHIND BAR · JOINS ISLAND TO LAND

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
SPIT → one end attached	LAGOON → water enclosed behind bar	SPIT → one end attached
BAR → crosses bay or mouth	TOMBOLO → joins island to land	BAR → crosses bay or mouth

CORE CLAIM
• SPIT → one end attached
• BAR → crosses bay or mouth

MECHANISM
• LAGOON → water enclosed behind bar
• TOMBOLO → joins island to land

CONSEQUENCE / LIMIT
• SPIT → one end attached
• BAR → crosses bay or mouth

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LAGOON → water enclosed behind bar; TOMBOLO → joins island to land.

05

STAGE 05

RELATIVE SEA-LEVEL FORK

RELATIVE SEA-LEVEL FORK · LAND RISE · SEA FALL · RAISED BEACH + TERRACE · EXPOSED MARINE FORMS · LAND FALL · SEA RISE · DROWNED RIVER VALLEY

LAND RISE / SEA FALL → emergence

RAISED BEACH + TERRACE → exposed marine forms → LAND FALL / SEA RISE → submergence → RIA = drowned river valley / FJORD = glacial trough

CORE CLAIM
• LAND RISE / SEA FALL → emergence
• RAISED BEACH + TERRACE → exposed marine forms

MECHANISM
• LAND FALL / SEA RISE → submergence
• RIA = drowned river valley / FJORD = glacial trough

CONSEQUENCE / LIMIT
• LAND RISE / SEA FALL → emergence
• RAISED BEACH + TERRACE → exposed marine forms

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RIA = drowned river valley / FJORD = glacial trough.

06

STAGE 06

STORM-SURGE CHAIN

STORM-SURGE CHAIN · CYCLONE WIND + LOW PRESSURE · WATER SETUP · TRACK + SHELF + BATHYMETRY · LOCAL AMPLIFICATION · HIGH TIDE + WAVES + RIVER FLOW · COMPOUND LEVEL · EXPOSURE + WEAK EVACUATION

CYCLONE WIND + LOW PRESSURE → water setup → TRACK + SHELF + BATHYMETRY → local amplification → HIGH TIDE + WAVES + RIVER FLOW → compound level → EXPOSURE + WEAK EVACUATION → disaster risk

CORE CLAIM
• CYCLONE WIND + LOW PRESSURE → water setup
• TRACK + SHELF + BATHYMETRY → local amplification

MECHANISM
• HIGH TIDE + WAVES + RIVER FLOW → compound level
• EXPOSURE + WEAK EVACUATION → disaster risk

CONSEQUENCE / LIMIT
• CYCLONE WIND + LOW PRESSURE → water setup
• TRACK + SHELF + BATHYMETRY → local amplification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HIGH TIDE + WAVES + RIVER FLOW → compound level.

STAGE 06 STORM-SURGE CHAIN

STORM-SURGE CHAIN · CYCLONE WIND + LOW PRESSURE · WATER SETUP · TRACK + SHELF + BATHYMETRY · LOCAL AMPLIFICATION · HIGH TIDE + WAVES + RIVER FLOW · COMPOUND LEVEL · EXPOSURE + WEAK EVACUATION

CYCLONE WIND + LOW PRESSURE → water setup → TRACK + SHELF + BATHYMETRY → local amplification → HIGH TIDE + WAVES + RIVER FLOW → compound level → EXPOSURE + WEAK EVACUATION → disaster risk

CORE CLAIM

- CYCLONE WIND + LOW PRESSURE → water setup
- TRACK + SHELF + BATHYMETRY → local amplification

MECHANISM

- HIGH TIDE + WAVES + RIVER FLOW → compound level
- EXPOSURE + WEAK EVACUATION → disaster risk

CONSEQUENCE / LIMIT

- CYCLONE WIND + LOW PRESSURE → water setup
- TRACK + SHELF + BATHYMETRY → local amplification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HIGH TIDE + WAVES + RIVER FLOW → compound level.

STAGE 07 INDIA COAST CONTRAST

INDIA COAST CONTRAST · BROADER PLAINS + LARGE DELTAS · NARROWER PLAIN + MANY ESTUARIES · PORT TEST · DEPTH, SHELTER, SILTATION, ENGINEERING · TENDENCY, NOT UNIVERSAL BINARY

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
EAST → broader plains + large deltas	PORT TEST → depth, shelter, siltation, engineering	EAST → broader plains + large deltas
WEST → narrower plain + many estuaries	RULE → tendency, not universal binary	WEST → narrower plain + many estuaries

CORE CLAIM

- EAST → broader plains + large deltas
- WEST → narrower plain + many estuaries

MECHANISM

- PORT TEST → depth, shelter, siltation, engineering
- RULE → tendency, not universal binary

CONSEQUENCE / LIMIT

- EAST → broader plains + large deltas
- WEST → narrower plain + many estuaries

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PORT TEST → depth, shelter, siltation, engineering; RULE → tendency, not universal binary.

STAGE 08 NCCR EVIDENCE BOUNDARY

NCCR EVIDENCE BOUNDARY · 1990-2018 ASSESSMENT · ABOUT 33.6 PERCENT · ABOUT 26.9 PERCENT · ABOUT 39.6 PERCENT; NOT AN ANNUAL RATE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
PERIOD → 1990-2018 assessment	ACCRETING → about 26.9 percent	PERIOD → 1990-2018 assessment
ERODING → about 33.6 percent	STABLE → about 39.6 percent; not an annual rate	ERODING → about 33.6 percent

CORE CLAIM

- PERIOD → 1990-2018 assessment
- ERODING → about 33.6 percent

MECHANISM

- ACCRETING → about 26.9 percent
- STABLE → about 39.6 percent; not an annual rate

CONSEQUENCE / LIMIT

- PERIOD → 1990-2018 assessment
- ERODING → about 33.6 percent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STABLE → about 39.6 percent; not an annual rate.

STAGE 09 CRZ IMPLEMENTATION LADDER

CRZ IMPLEMENTATION LADDER · CATEGORY-SPECIFIC NATIONAL RULES · HTL/LTL AND TECHNICAL INPUTS · CONSULTATION + APPROVAL + LOCAL MAP · APPRAISAL, DECISION, COMPLIANCE · CLOSE DISTINCTION · EMERGENT AND SUBMERGENT COASTS DESCRIBE RELATIVE SEA-LEVEL HISTORIES, EROSIONAL

NOTIFICATION → category-specific national rules

MAPPING → HTL/LTL and technical inputs → CZMP → consultation + approval + local map → PROJECT → appraisal, decision, compliance

CORE CLAIM

- NOTIFICATION → category-specific national rules
- MAPPING → HTL/LTL and technical inputs

MECHANISM

- CZMP → consultation + approval + local map
- PROJECT → appraisal, decision, compliance

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Emergent and submergent coasts describe relative sea-level histories, erosional and depositional forms
- can coexist, and CRZ categories/regulatory lines are legal-planning constructs rather than geomorphic shoreline types.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Emergent and submergent coasts describe relative sea-level histories, erosional and depositional forms can coexist, and CRZ categories/regulatory lines are legal-planning constructs rather than geomorphic shoreline types.

MAPPING → HTL/LTL and technical inputs → CZMP → consultation + approval + local map → PROJECT → appraisal, decision, compliance

CORE CLAIM
- NOTIFICATION → category-specific national rules
- MAPPING → HTL/LTL and technical inputs

MECHANISM
- CZMP → consultation + approval + local map
- PROJECT → appraisal, decision, compliance

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Emergent and submergent coasts describe relative sea-level histories, erosional and depositional forms
- can coexist, and CRZ categories/regulatory lines are legal-planning constructs rather than geomorphic shoreline types.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Emergent and submergent coasts describe relative sea-level histories, erosional and depositional forms can coexist, and CRZ categories/regulatory lines are legal-planning constructs rather than geomorphic shoreline types.

10

STAGE 10 COASTAL RESPONSE HIERARCHY

COASTAL RESPONSE HIERARCHY | KEEP NEW EXPOSURE OUT OF HIGH-RISK SPACE | BUFFERS + RESILIENT USE | MOVE ASSETS WHERE PERSISTENCE FAILS | SELECTIVE WORKS WITH DOWNDRIFT AUDIT | EVIDENCE LIMIT | EROSION CONTROL CAN SHIFT RISK DOWNDRIFT; PORTS, DAMS, SAND

AVOID → keep new exposure out of high-risk space

RESTORE / ACCOMMODATE → buffers + resilient use → RETREAT → move assets where persistence fails → PROTECT → selective works with downdrift audit

CORE CLAIM
- AVOID → keep new exposure out of high-risk space
- RESTORE / ACCOMMODATE → buffers + resilient use

MECHANISM
- RETREAT → move assets where persistence fails
- PROTECT → selective works with downdrift audit

CONSEQUENCE / LIMIT
- EVIDENCE LIMIT: Erosion control can shift risk downdrift
- ports, dams, sand mining, storms, subsidence and sea-level rise interact, while CRZ rules, clearances and project status require

MECHANISM / VERDICT — EVIDENCE LIMIT: Erosion control can shift risk downdrift; ports, dams, sand mining, storms, subsidence and sea-level rise interact, while CRZ rules, clearances and project status require exact notification/date/source discipline.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Erosion control can shift risk downdrift; ports, dams, sand mining, storms, subsidence and sea-level rise interact, while CRZ rules, clearances and project status require exact notification/date/source discipline.

11

STAGE 11 COAST ANSWER SPINE

COAST ANSWER SPINE | DEFINE + MAP | PROCESS AND COASTAL SECTOR | ENERGY, SEDIMENT AND SEA-LEVEL CHAIN | PERIOD, CATEGORY, PLAN AND STATUS | INTEGRATED, ADAPTIVE AND LOCALLY MAPPED

DEFINE + MAP → process and coastal sector → DERIVE → energy, sediment and sea-level chain → VERIFY → period, category, plan and status → VERDICT → integrated, adaptive and locally mapped

CORE CLAIM
- DEFINE + MAP → process and coastal sector
- DERIVE → energy, sediment and sea-level chain

MECHANISM
- VERIFY → period, category, plan and status
- VERDICT → integrated, adaptive and locally mapped

CONSEQUENCE / LIMIT
- DEFINE + MAP → process and coastal sector
- DERIVE → energy, sediment and sea-level chain

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DEFINE + MAP → process and coastal sector.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | D.R. KHULLAR LISTS CYCLONE DANGERS AS GALES/STRONG WINDS, TORRENTIAL | KHULLAR NOTES MOST CASUALTIES FROM CYCLONES ARE CAUSED BY | TIDAL WAVES/STORM SURGES, WITH SEVERE SURGES PENETRATING INLAND FROM | LOW PRESSURE RAISES SEA LEVEL SLIGHTLY, BUT THE MAIN | IS STRONG ONSHORE CYCLONE WINDS PILING SEAWATER AGAINST A | INDIA LINK | THE SHALLOW, FUNNEL-SHAPED BAY OF BENGAL AND DELTAIC EAST

OPTIONAL SOURCE DEPTH
- D.R. Khullar lists cyclone dangers as gales/strong winds, torrential rain and high tidal waves, also called storm surges .
- Khullar notes most casualties from cyclones are caused by coastal inundation by
- tidal waves/storm surges, with severe surges penetrating inland from the coast.
- CAUTION: Mechanism: low pressure raises sea level slightly, but the main driver

ADVANCED DEBATE
- is strong onshore cyclone winds piling seawater against a shallow coast.
- CAUTION: India link: the shallow, funnel-shaped Bay of Bengal and deltaic east coast make Odisha-West Bengal-Bangladesh especially surge-prone.
- Majid Husain describes tropical muddy coasts with mangroves in sediment-rich lagoons, bays, deltas and estuaries, and also mentions salt marshes.
- D.R. Khullar notes mangroves occur along India's sheltered estuaries, tidal creeks, backwaters, salt marshes and mudflats.

LEGACY / NUANCE
- CAUTION: Blue carbon = carbon stored by coastal and marine ecosystems,
- especially mangroves, seagrasses and salt marshes, in biomass and waterlogged sediments.
- CAUTION: India link: Sundarbans, Mahanadi-Godavari-Krishna deltas, Gulf of Kachchh, Gulf of
- Mannar and Andaman-Nicobar ecosystems link coastal protection with climate mitigation.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels